YANHUI YANG

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EDUCATION

PhD Program in Physics University of California, Riverside — Sep. 2022–present
Concentration in Astronomy
Overall GPA: 4.00/4.00

BS in Astronomy University of Science and Technology of China — Jul. 2021
Wang Shouguan Talent Program in Astronomy
Thesis: *Hydrodynamic simulations of turbulent mixing layers of galactic winds*
Overall GPA: 3.74/4.30, Major GPA: 3.93/4.30 (Physics & Astronomy courses)

RESEARCH INTERESTS

Cosmology: large-scale structure, dark matter, dark energy, cosmological emulation; Astrophysics: galaxy formation and evolution, circumgalactic medium, turbulent mixing layers

APPOINTMENTS

Graduate Student Researcher, Advisor: Simeon Bird (UCR) — Fall 2023–present
Teaching Assistant for *General Physics Laboratory 2LC*, Coordinator: Jonathan Eldridge (UCR) — Winter–Spring 2023
Research Assistant, Advisor: Suoqing Ji (SHAO) — Spring 2022
Summer Research Student, Philip F. Hopkins's group (Caltech, remote); Research mentor: Suoqing Ji — Summer 2020

PUBLICATIONS

Yanhui Yang, Simeon Bird, Yihao Zhou, Tiziana Di Matteo, Rupert Croft, Yueying Ni, and Nianyi Chen. Matter Clustering in Astrid: Reduced Baryonic Suppression from Realistic Black Hole Dynamics. *ApJL*, 1007(1):L20, August 2026. doi: 10.3847/2041-8213/ae8dc6.

Yanhui Yang, Simeon Bird, Ming-Feng Ho, and Mahdi Qezlou. Ten-dimensional neural network emulator for the nonlinear matter power spectrum. *Phys. Rev. Lett.*, 136(6):061001, February 2026a. doi: 10.1103/8yzy-t59w.

Yanhui Yang, Simeon Bird, Ming-Feng Ho, and Mahdi Qezlou. Design and optimization of neural networks for multifidelity cosmological emulation. *Phys. Rev. D*, 113(4):043508, February 2026b. doi: 10.1103/cqrc-k8wq.

Yanhui Yang, Simeon Bird, and Ming-Feng Ho. Ten-parameter simulation suite for cosmological emulation beyond ΛCDM. *Phys. Rev. D*, 111(8):083529, April 2025. doi: 10.1103/PhysRevD.111.083529.

Yanhui Yang and Suoqing Ji. Radiative turbulent mixing layers at high Mach numbers. *MNRAS*, 520(2):2148–2162, April 2023. doi: 10.1093/mnras/stad264.

Only first-author papers are listed above. Please refer to NASA/ADS (ORCID: 0000-0001-6221-6024) for a complete list of my publications.

PRESENTATIONS

Invited Talks

Weak Baryonic Suppression in Astrid: A Challenge for Precision Cosmology
University of Kansas Astronomy & Space Physics Seminar, The University of Kansas — Sep. 2026

Modeling Cosmic Large-Scale Structure Beyond ΛCDM with Simulations and Emulation
Fudan-CfAA Seminar, Fudan University — Oct. 2025

Goku: A 10-parameter simulation suite for cosmological emulation
Physics and Astronomy Student Seminar, University of California, Riverside — Mar. 2025

Radiative turbulent mixing layers at high Mach numbers
Shanghai Astronomical Observatory, Chinese Academy of Sciences — May 2022

Contributed Talks and Posters

Weak Baryonic Suppression in Astrid: A Challenge for Precision Cosmology [video]
Cosmology from Home 2026 — Sep. 2026

Cosmological Emulation based on the Goku Simulations in a 10-Dimensional Parameter Space
XIX International Conference on Interconnections between Particle Physics and Cosmology (PPC2026)
The University of New South Wales, Sydney — Aug. 2026

Weak Baryonic Suppression in Astrid: A Challenge for Roman Weak Lensing Cosmology
Shifting Landscapes in Astrophysics: New Frontiers to Explore with Roman
California Institute of Technology/IPAC — Jul. 2026

Improving Cosmological Emulation via Multifidelity Neural Networks
Galaxy Formation & Evolution in Southern California, University of California, San Diego — Sep. 2025

Nonlinear Matter Power Spectrum Emulation beyond ΛCDM for Next-Generation Surveys (poster)
Cosmic Cartography with Roman 2025, STScI, Baltimore, MD — Jul. 2025

10D emulation for the nonlinear matter power spectrum beyond ΛCDM based on Goku [video]
Cosmology from Home 2025 — Jul. 2025

Goku: A 10-parameter simulation suite for cosmological emulation
Frontera User Meeting 2024, TACC, The University of Texas at Austin — Aug. 2024

ADVISORS & REFERENCES

Simeon Bird (sbird@ucr.edu) — PhD advisor
Associate Professor, Department of Physics and Astronomy, University of California, Riverside

Tiziana Di Matteo (tiziana@phys.cmu.edu)
Professor, Department of Physics, Carnegie Mellon University
Director of the McWilliams Center for Cosmology and Astrophysics

Rupert A. C. Croft (rcroft@cmu.edu)
Professor, Department of Physics, Carnegie Mellon University

Suoqing Ji (sqji@fudan.edu.cn)
Associate Professor, Department of Physics, Fudan University

Xiaobo Dong (xbdong@ynao.ac.cn)
Research Professor, Yunnan Observatories, Chinese Academy of Sciences

AWARDS & HONORS

Dean's Distinguished Fellowship (University of California, Riverside) — 2022

NAOC Scholarship (National Astronomical Observatories, Chinese Academy of Sciences) — 2020

National Encouragement Scholarship — 2020

Outstanding Student Scholarship (Grade 3) — 2019

Zhang Zongzhi Sci-Tech Scholarship — 2018

MENTORSHIP & SUPERVISION

Undergraduate students (co-mentored with Prof. Simeon Bird): Mark Achenbach (UH Mānoa), Michael Padilla (UCR), Mohak Bhattacharya (UCR)

PROFESSIONAL SERVICE

Journal referee for *Physical Review D* and *Physical Review Letters*

SKILLS

Programming languages	C (GSL, MPICH, OpenMP), C++ (Boost, CUDA), Python (PyTorch, mpi4py, PyMC, pandas, NumPy, Matplotlib, yt, BigFile, GPy, etc.), Bash, Fortran, HTML, CSS, Markdown, LaTeX
Research software	MP-Gadget, CLASS, FLASH, GADGET-3, RAMSES, AREPO
Tools	Git, Mathematica, Origin, Gnuplot, IDL, DS9, VS Code, Codex

VISITS & ADDITIONAL TRAINING

TACC Open Hackathon, Texas Advanced Computing Center, UT Austin	Oct. 2024
The 2023 Machine Learning Institute, TACC, UT Austin	Jun. 2023
Shanghai Astronomical Observatory, Chinese Academy of Sciences	Jan. 2020
Five-hundred-meter Aperture Spherical Telescope (FAST)	Jul. 2019

OUTREACH

Physics Fun Expedition: A Science Outreach Program for Township Middle Schools	Jul. 2019